



East West University

An Evaluation of Long-run Economic Growth: Testing the Solow Growth Model

Prepared for


Senior Lecturer
Department of Economics
East West University

Prepared By

Irfan Ahmed
Bachelors in Social Science (Economics)

Introduction

Economic prosperity depends on production, growth, and development, requiring an examination of long-term economic growth models like the Solow growth model to assess a country's trajectory. Economic growth, both short-term (real GDP increase) and long-term (potential GDP growth), reflects the production of output over time and the development potential of the economy. Long-term economic growth involves a sustained increase in the production of goods and services.

The Solow Growth model explains how capital accumulation influences long-term economic growth. In Bangladesh and Malaysia, sustained economic growth requires both capital and labor to increase, with capital growing at a faster rate than labor to maintain a high capital-labor ratio. However, as the capital-labor ratio rises, productivity per worker declines, leading to a decrease in national income, savings, investment, and capital formation. This process continues until the growth rates of capital and labor equalize, achieving a balance in the economy's production factors. This study aims to test the applicability of the Solow growth model in assessing the long-term economic growth of Bangladesh and Malaysia.

First, the fundamental theoretical foundation of the Solow growth model will be described. Next, the data and variables used will be explained. Finally, an empirical analysis, estimation, and comparison of the TFP growth rate, indicators that explain the TFP growth rate and the convergence hypothesis will be attempted to draw a conclusion.

Theoretical Framework

Solow growth Model: is shown by the given saving, investment and consumption rate, how the accumulation of capital leads to the per capita GDP growth in the long-run. The production function of Solow growth model:

$$\begin{aligned} Y &= f(K, L) \\ &= AK^{1/3} L^{2/3} \end{aligned}$$

Here,

A=Technical coefficient or Total factor Productivity

K=Capital

L=Labor

(1/3) = Output elasticity of Capital

(2/3) = Output elasticity of Labor

Assumptions:

Y_t can be divided into two parts

$Y_t = C_t + I_t$ (closed economy)

Since the capital will be accumulated over time, the capital accumulation function is:

$$K_{t+1} = K_t + I_t - dK_t$$

Capital accumulation in future depends on the initial stock of capital and net investment at t .

$$K_{t+1} = K_t + I_t - dK_t$$

$$\Rightarrow K_{t+1} - K_t = I_t - dK_t$$

$$\Rightarrow \Delta K = I - dK_t$$

Changes in capital depend on net investment.

Solow growth model includes 5 equations and 5 unknown variables:

1. Production function: $y = A t^{1/3} K^{2/3} L t^{2/3}$
2. Capital accumulation: $\Delta K_{t+1} = I_t - d K_t$
3. Resource constraint: $Y_t = C_t + I_t$
4. Labor force: $L_t = L$
5. Allocation of resources: $I_t = s Y_t$

Unknown variables: Y_t, K_t, L_t, C_t, I_t .

Exogenous variables: $\underline{Y_t}, \underline{K_t}, \underline{Y_t}, \underline{L_t}, \underline{L}$

Steady State Condition:

A characteristic of a condition, such as value, rate, periodicity, or amplitude, exhibiting only negligible change over an arbitrarily long period of time.

Now, we can get the steady state capital when the net investment equals to 0 at time period t . So, we can express the steady state condition:

$$\Delta K_{t+1} = 0$$

$$\Rightarrow I_t - dK_t = 0 \text{ (i)}$$

We know the investment function: $I_t = sY_t$

Putting this equation in (i),

$$sY_t - dK_t = 0$$

$$=s(AK^{1/3} L^{2/3}) - dK_t = 0$$

By mathematical calculation, we will get,

$$K_t = (sA/d)^{3/2} L$$

This is the steady-state solution of capital. Here, A and L are considered to be fixed over time. So, capital accumulation depends on the value of s .

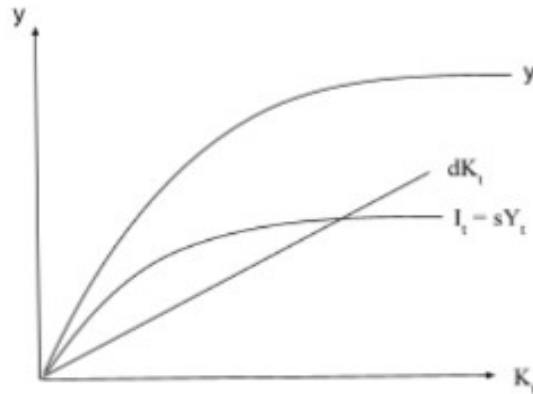


Fig: Solow Growth Model

This graph shows how capital will accumulate over time ($t \rightarrow \infty$) with the starting initial capital K_0 . So, when savings/investment equals the depreciation rate, the economy will reach a steady-state condition.

Data and variables:

In this term paper we use data from 1970 to 2020 for Bangladesh and Malaysia. The equation used in this term paper are mentioned below:

- $y = f(K, L) = AK^{1/3} L^{2/3}$

For empirical analysis we take variables:

- Per capita real GDP = GDP of the country/ Population of that country

The Solow growth model focuses on long-run economic growth. The growth in real GDP per capita indicates the pace of income growth per head of the population. As a single composite indicator, it is a powerful summary indicator of economic development. So, we take this variable for our analysis.

- Real Capital output ratio = $\Delta K / \Delta y$

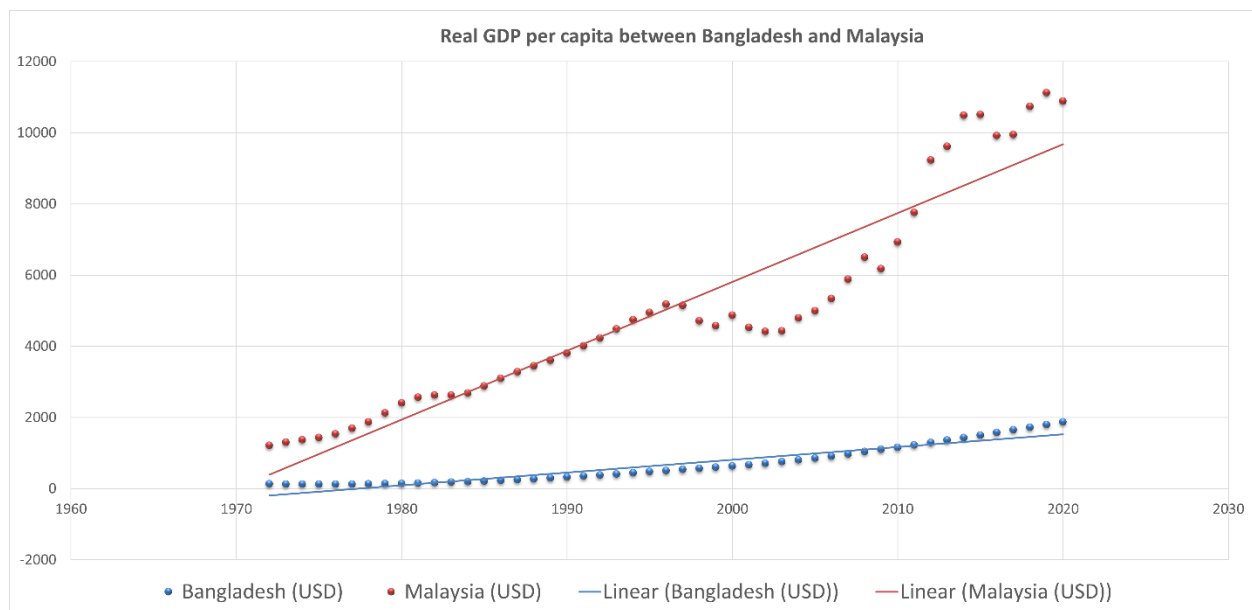
- Average TFP growth rate $g = (A_t)^{1/t} - 1$.

The TFP gives us the growth in real output, which is in addition to the growth in inputs such as labor or capital employed for productive purposes. The Solow growth model represents output growth that happens beyond the simple growth of inputs. As such, the Solow growth model is often described as a measure of productivity growth due to technological innovation. The Solow residual is also referred to as total factor productivity. The other variables are GDP at constant, Labor information, Population, Deflator, Capital Stock, HDI.

[The data has been collected from “APO Productivity Database 2020; The Worldwide Governance Indicators, 2020 Update”.]

Empirical Analysis

Differences between Bangladesh and Malaysia can be analyzed using real GDP per capita, which measures output per person adjusted for inflation. The scatter graph below shows the disparity in per capita real GDP between the two countries from 1972 to 2020.



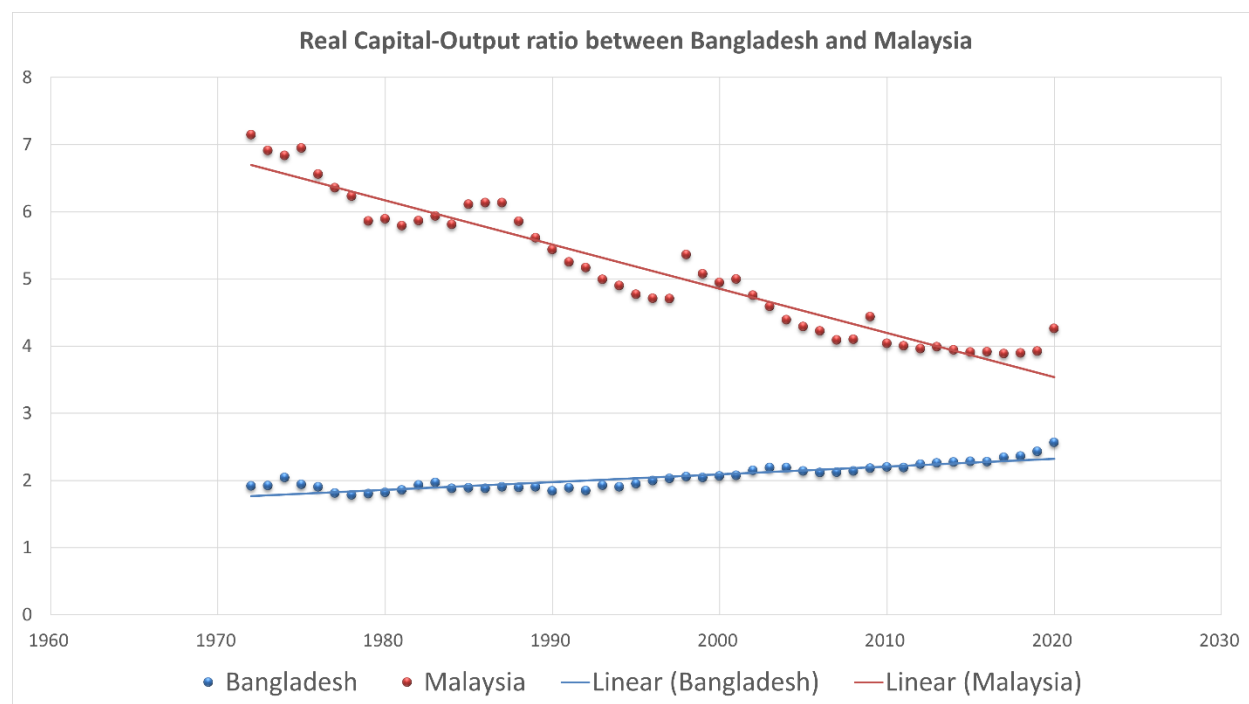
The graph above helps us understand the differences in per capita GDP between Bangladesh and Malaysia over the years, relating to the Solow growth model. Factors contributing to these differences can be summarized as follows:

Capital accumulation: Malaysia consistently outperforms Bangladesh in per capita GDP due to higher levels of investment in physical capital, including infrastructure, machinery, and technology. This capital accumulation contributes to Malaysia's higher productivity and output per capita.

Technological progress: Malaysia's adoption of advanced technologies and innovation plays a significant role in its higher per capita GDP. Technological advancements increase productivity and drive economic growth, giving Malaysia an edge over Bangladesh.

Labor force: While Bangladesh has a larger population, Malaysia surpasses it in per capita GDP. This can be attributed to Malaysia's investment in education and skill development, resulting in a higher quality labor force. Despite having a smaller labor force, Malaysia's higher productivity leads to higher per capita GDP.

Real capital-output ratio for sample countries (1972 -2020)



The real capital-output ratio measures the amount of capital (machinery, equipment, infrastructure, etc.) required to produce a unit of output (GDP).

- Bangladesh: The real capital-output ratio in Bangladesh ranges from 1.783 to 2.563. The ratio indicates that Bangladesh has been relatively capital-intensive in its production processes compared to Malaysia. The capital-output ratio in Bangladesh has generally been higher than Malaysia, except in 2014 and 2015. Bangladesh's capital-output ratio experienced fluctuations over time, but overall, it has shown a slight increasing trend.

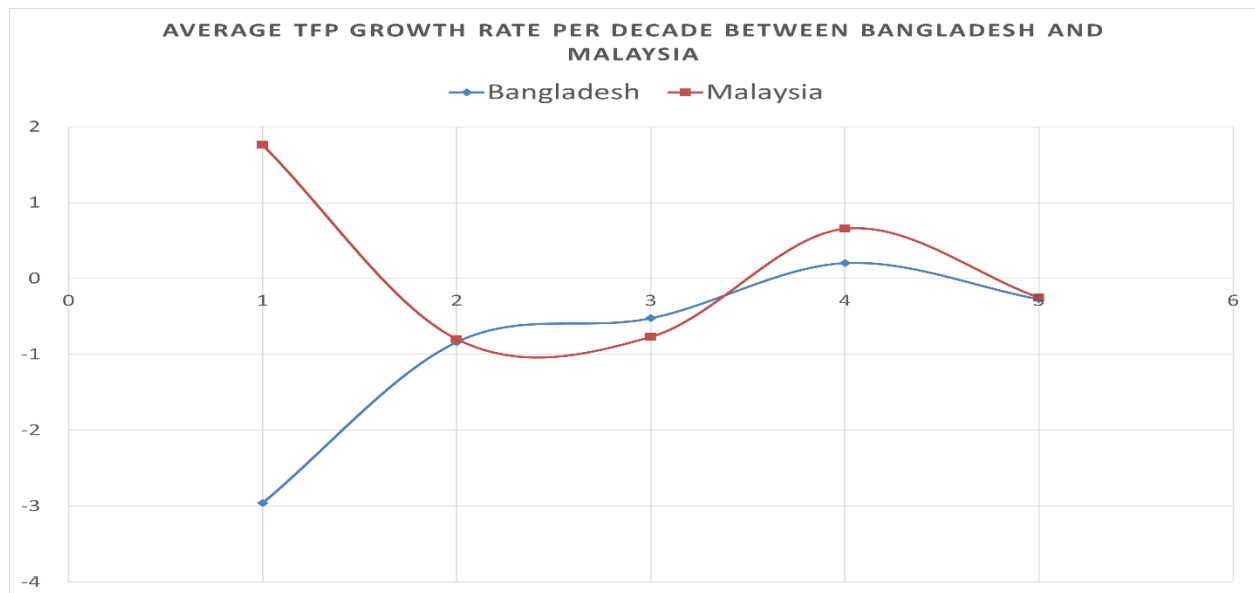
- Malaysia: The real capital-output ratio in Malaysia remains constant at 4 for most of the years. Malaysia has a consistently lower capital-output ratio compared to Bangladesh. A lower capital-output ratio suggests that Malaysia has been able to achieve higher output (GDP) with relatively less capital investment compared to Bangladesh. Malaysia's capital-output ratio has remained stable over the years, indicating a consistent level of efficiency in its production processes.

Overall, the data suggests that Bangladesh has had a higher capital-output ratio and has been more capital-intensive in its production processes compared to Malaysia. This indicates that Bangladesh requires a relatively higher amount of capital investment to produce a unit of output compared to Malaysia. On the other hand, Malaysia has shown a lower capital-output ratio, implying a more efficient utilization of capital in its production processes.

Average TFP growth rate per decade between Bangladesh and Malaysia

Furthermore, the analysis of Total Factor Productivity (TFP) can also aid as a valuable tool in the evaluation of the two economies. TFP is a measure of the efficiency and effectiveness with which inputs (such as labor and capital) are transformed into output. It reflects technological progress, innovation, managerial skills, and institutional factors that contribute to the overall productivity of an economy.

The graph below shows the average TFP growth rate per decade (from 1970s to 2010s), between Bangladesh and Malaysia.



The graph provides the average total factor productivity (TFP) growth rates for Bangladesh and Malaysia over different decades, ranging from 1970s, 1980s, 1990s, 2000s and 2010s. With the assessment of the scatter graph above, following observations can be made.

Based on the TFP growth data, Malaysia demonstrated significantly higher TFP growth compared to Bangladesh in the 1970s. In the 1980s, both countries experienced negative TFP growth, but the difference between their values was not substantial. Similarly, in the 1990s, both countries had negative TFP growth, with Malaysia slightly surpassing Bangladesh in the decline. However, in the 2000s, both countries achieved positive TFP growth, with Malaysia exhibiting a higher growth rate compared to Bangladesh. Lastly, in the 2010s, both countries experienced negative TFP growth, but the disparity between their values was minimal.

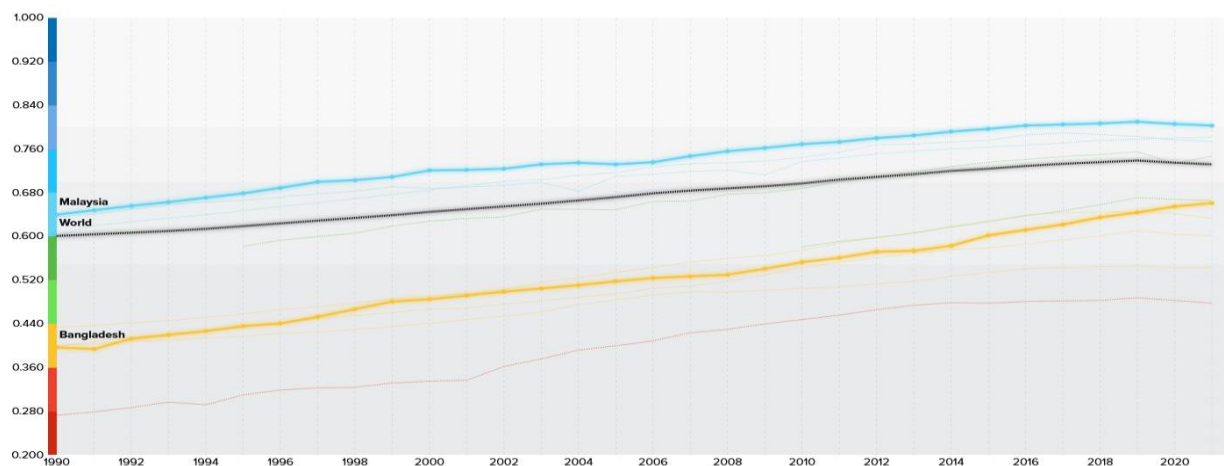
Significance of TFP for Higher Per Capita Income

Total Factor Productivity (TFP) is crucial for economic growth and higher per capita income. Higher TFP allows countries to produce more output with the same inputs, leading to increased efficiency and productivity. This results in higher economic growth and per capita income.

Higher TFP levels also enhance a country's competitiveness in the global market. Improved productivity and efficiency enable nations to produce goods and services at lower costs, making them more competitive in international trade. This leads to increased exports and higher revenues for domestic producers, positively impacting per capita income.

Additionally, higher TFP growth contributes to higher wages and incomes for workers. Improved productivity allows businesses to pay higher wages to attract and retain skilled workers, ultimately improving the standard of living and increasing per capita income.

Lastly, TFP growth is a sustainable driver of economic growth. Unlike factors such as capital accumulation, TFP growth can continue to foster innovation, technological progress, and efficiency gains, resulting in sustainable increases in per capita income.



This can also be evidenced with the data above, as it can be seen that Malaysia consistently had a higher Human Development Index (HDI) compared to Bangladesh. In most cases, a higher HDI means that the productive potential of the factor is higher. Therefore, a higher HDI can be linked with higher TFP growth as well.

Further evidence: include the average years of schooling that children receive. In Bangladesh, it is 7 years; whereas, in Malaysia they receive 11 years of schooling on average. Additionally, other

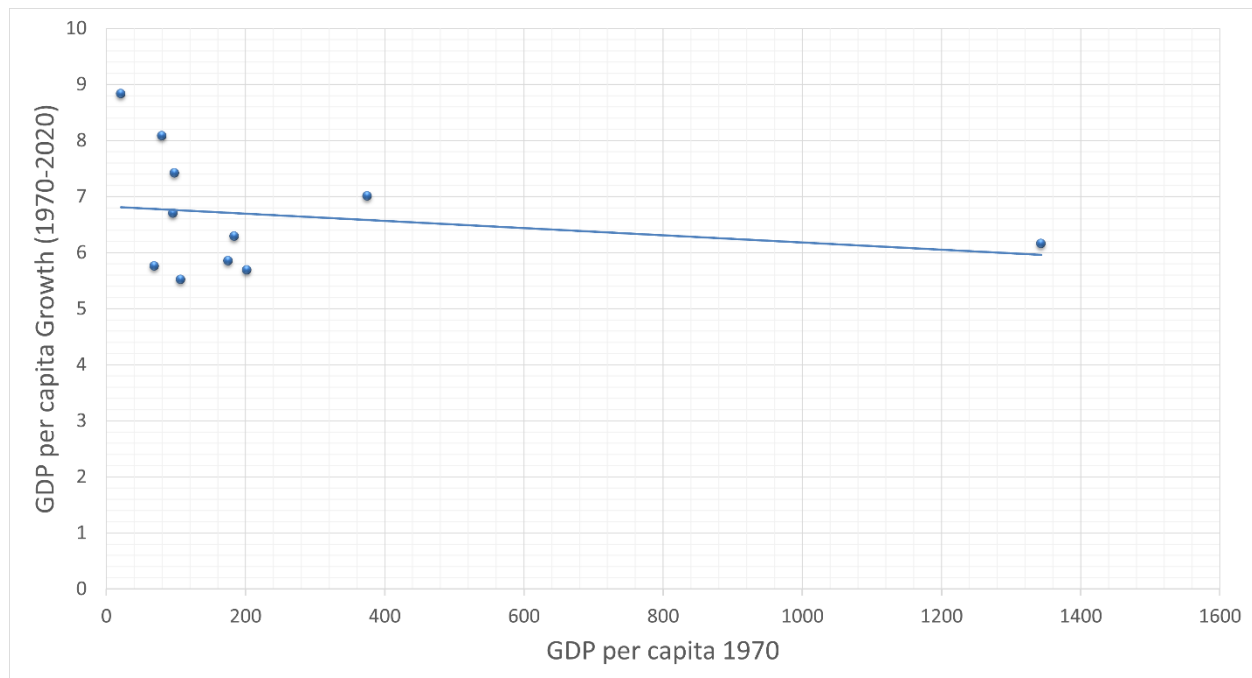
indicators could be the Rule of Law index, where Bangladesh resides at 38.8 and Malaysia boasts a score of 65.6 – which shows that Malaysia has stronger institutional enforcement and better control of their systems compared to Bangladesh. Both of these indicators (at a higher mark) could signify that the economy is likely to experience higher TFP growth.

Convergence Test

Consider the following chart.

Country	GDP per Capita (1970)	Growth (1970-2020) (%)
Bangladesh	96	6.7
Indonesia	80	8.08
Sir Lanka	184	6.29
Philippines	202	5.69
Myanmar	21	8.83
Malaysia	375	7.01
Vietnam	98	7.42
Cambodia	107	5.52
Bhutan	175	5.85
Nepal	69	5.76
Brunei	1343	6.16

If we plot this table against GDP per capita from 1970 (in x-axis) and Growth rate (in y-axis) over the 50 years, we get the following graph.



With the combination of the graph and table above, we can make an assessment for the convergence hypothesis: countries which had lower GDP per capita in 1970 experienced higher growth over the years, which means they are moving towards the steady-state condition and countries which had higher GDP per capita in 1970 experienced comparatively lower growth, which indicates that they have already moved closer to their steady-state condition. Therefore, proving the convergence hypothesis.